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OepnCV Image, Video, Cam Examples

To know how to use OpenCV, it is easier to see examples than to know each function.

Python and C++ are similar. If you know one language, it is easy to learn the other; compare the Python and C++ code that performs the same.

Image

C++

Python

image_show.cpp

```
#include <opencv2/opencv.hpp>

int main( void )
{
    cv::Mat img;

    /// Reads image.jpg and assigns it in the img variable.
    img = cv::imread( "image.jpg" );

    /// Shows the img on the image window.
    cv::imshow( "image", img );

    /// Waits for a key event infinitely.
    cv::waitKey();
    cv::destroyAllWindows();

    return 0;
}
```

```
$ g++ image_show.cpp -o image_show $(pkg-config opencv4 --libs --cf
lags)
```

```
$ ./image_show
```

Video

C++

Python

video_show.cpp

```
#include <opencv2/opencv.hpp>

int main( void )
{
    cv::Mat frame;

    /// Opens video.mp4.
    cv::VideoCapture cap;
    cap.open( "video.mp4" );

    /// Gets frames per second.
    int fps = cap.get( cv::CAP_PROP_FPS );

    if( cap.isOpened() )
    {
        for( ;; )
        {
            /// Reads a frame and assigns it in to the frame variable.
            cap >> frame;
            if( frame.empty() )
            {
                break;
            }

            /// Shows the frame on the video window.
            cv::imshow( "video", frame );

            /// Waits for a key event for 1000/fps ms.
            /// If the key event occurs, it breaks the loop.
            if( ( cv::waitKey( 1000 / fps ) & 0xFF ) != 0xFF )
            {
                break;
            }
        }
    }

    cap.release();
    cv::destroyAllWindows();

    return 0;
}
```

```
$ g++ video_show.cpp -o video_show $(pkg-config opencv4 --libs --cf  
lags)  
  
$ ./video_show
```

Cam recording

Before you start, check your cam info using the command below.

```
$ v4l2-ctl --all --list-formats-ext
```

If you want to know specific cam device file like `/dev/video0`.
Execute the command `v4l2-ctl -all -list-formats-ext -d /dev/video0`.

C++

Python

cam_record.cpp

```
#include <opencv2/opencv.hpp>

const int OUTPUT_VIDEO_WIDTH  = 640;
const int OUTPUT_VIDEO_HEIGHT = 480;

int main( void )
{
    cv::Mat frame;

    /// Opens the cam number 0. (/dev/video0)
    cv::VideoCapture cap;
    cap.open( 0, cv::CAP_V4L2 );

    /// Sets the width and height in pixel.
    cap.set( cv::CAP_PROP_FRAME_WIDTH, OUTPUT_VIDEO_WIDTH );
    cap.set( cv::CAP_PROP_FRAME_HEIGHT, OUTPUT_VIDEO_HEIGHT );

    /// Sets the format of pixel.
    cap.set( cv::CAP_PROP_FOURCC, cv::VideoWriter::fourcc( 'M',
'J', 'P', 'G' ) );

    /// Sets the recording properties.
    cv::VideoWriter out;
    /// Saves as record.avi and uses FFmpeg.
    out.open( "record.avi", cv::VideoWriter::fourcc( 'F', 'M', 'P',
'4' ), 25.0,
                cv::Size( OUTPUT_VIDEO_WIDTH, OUTPUT_VIDEO_HEIGHT )
    );

    if( cap.isOpened() && out.isOpened() )
    {
        for( ;; )
        {
            try
            {
                /// Reads a frame and assigns it in the frame variable.
                cap.read( frame );
            }
            catch( cv::Exception& e )
            {
                continue;
            }

            if( frame.empty() )
            {
                break;
            }
        }
    }
}
```

```
    /// Writes the frame to record.avi.
    out.write( frame );

    /// Shows the frame on the video window.
    cv::imshow( "video", frame );

    /// Waits for a key event for 1 ms.
    /// If the key event occurs, it breaks the loop.
    if( ( cv::waitKey( 1 ) & 0xFF ) != 0xFF )
    {
        break;
    }
}

cap.release();
out.release();
cv::destroyAllWindows();

return 0;
}
```

```
$ g++ cam_record.cpp -o cam_record $(pkg-config opencv4 --libs --cf
lags)

$ ./cam_record
```

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